73. From January 1, 2021, the price of petrol (in Rupees per liter) on m^{th} day of the year is 80 + 0.1m, where m = 1, 2, 3 ....., 100 and thereafter remains constant. On the other hand, the price of diesel (in Rupees per litre) on n^{th} day of 2021 is 69 + 0.15n for any n. On which date in the year 2021 are the prices of these two fuels equal?

(a) 21st May (b) 20th May
(c) 19th May (d) 18th May

Ans. (b) : ∵ Price of petrol = 80 + 0.1 m
and m = 1, 2, 3 ……..100 and therefore
remains constant.
∴ Price of petrol after 100 days
= 80 + 0.1 × 100= 80 + 10 = ₹ 90
∵ Price of Diesel = 69 + 0.15 m
then price of Diesel after 100 days
= 69 + 0.15 × 100 = 69 + 15 = ₹ 84
After 100 days,
time taken the price of these two fuels equal

$$= \frac{₹(90-84)}{15 \text{ paise}} \Rightarrow \frac{6\times100}{15} \quad [\because ₹\,1 = 100 \text{ paise}] = 40 \text{ days.}$$

∴ total days = 100 + 40 = 140 days
∴ Required date = 1st of January 2021 + 140 days
= 20th of May 2021.

74. If 3^{2019} is divided by 10 then what is the remainder?

(a) 1 (b) 3
(c) 7 (d) 9

Ans. (c) :

$$\frac{3^{2019}}{10} = \frac{(3^2)^{1009}.3^1}{10} = \frac{9^{1009}.3}{10} = \frac{(-1)^{1009}.3}{10} = \frac{(-1\times3)}{10} = \frac{-3}{10}$$

∴ Remainder = 10 – 3 = 7

ENGLISH LANGUAGE & GRAMMAR COMPREHENSION

Directions for the following 4 (four) items :
Read the following four passages and answer the items that follow. Your answers to these items should be based on the passages only.

Passage – 1

Researchers simulated street lighting on artificial grassland plots containing pen-aphids, sap-sucking insects, at night. These were exposed to two different types of light—a white light similar to newer commercial LED lights and an amber light similar to sodium street lamps. The low intensity amber light was shown to inhibit, rather than induce, flowering in a wild plant of the pea family which is a sources of food for the pea-aphids in grasslands. The number of aphids was also significantly suppressed under the light treatment due to the limited amount of foods available.

75. Which one of the following statements best reflects the most critical inference that can be made from the passage given above?

(a) Low intensity light has more adverse effect on the plants as compared to high intensity light
(b) Light pollution can have a permanent adverse impact on ecosystem
(c) White light is better for the flowering of plants as compared to the light of other colours
(d) Proper intensity of light in an ecosystem is important not only for plants but for animals too

Ans. (d) : According to the given passage low-intensity light inhibits the flowering of pants. Further, it high lights that the number of aphids was also significantly suppressed under the light treatment due to the limited amount of food available. So, it can be inferred from the passage that the proper intensity of light in a ecosystem is important not only for plants (for flowering) but also for animals like aphid. Hence, option (d) is correct and rest of three are incorrect.

Passage – 2

Approximately 80 percent of all flowering plant species are pollinated by animals, including birds and mammals, but the main pollinators are insects. Pollination is responsible for providing us with a wide variety of food, as well as many plant-derived medicines. At least one-third of the world's agricultural crops depend upon pollination. Bees are the most dominant taxa when it comes to pollination and they are crucial to more than four hundred crops. Pollination is an essential service that is the result of intricate relationships between plants and animals, and the reduction or loss of either affects the survival of both. Effective pollination requires resources, such as refuges of pristine natural vegetation.

76. On the basis of the passage given above, the following assumptions have been made :

1. Sustainable production of India's cereal food grains is impossible without the diversity of pollinating animals
2. Monoculture of horticultural crops hampers the survival of insects
3. Pollinators become scarce in cultivated areas devoid of natural vegetation
4. Diversity in insects induces diversity of plants

Which of the above assumptions is/are valid?

(a) 1 only (b) 2, 3 and 4 only
(c) 1 and 2 only (d) 3 and 4 only

Ans. (d) : According to the above given lines of the passage "Pollination is an essential service that is the result of an intricate relationship between plant and animals, and the reduction or loss of either affects the survival of both. Effective Pollination requires resources, such as refuges of pristine natural vegetation", it can be assumed that the pollinators become scarce in cultivated areas devoid of natural vegetation and diversity in insects induced diversity of plants. Hence, option (d) is correct.

Passage – 3

A study conducted on the impacts of climate change over the Cauvery basin of Tamil Nadu using regional climate models showed an increasing trend for maximum and minimum temperatures, and a decrease in the number of rainy days. These climatic shifts will have an impact on the hydrological cycles in the region, lead to more run-off and less recharge, and effect the groundwater tables. Further, there has been and increase in the frequency of droughts in the State. This has driven farmers to increase dependency on groundwater resources to secure their crops.

77. Which one of the following statements *best reflects the crux* of the passage given above?

(a) Development of regional climate models helps in choosing climate-smart agricultural practices
(b) Heavy dependence on groundwater resources can be reduced by adopting dry-land cropping systems
(c) Climate changes increases the criticality of water resources while simultaneously threatening it
(d) Climate changes cause the farmers to adopt unsustainable livelihoods and risky coping strategies

Ans. (c) : The main point of the passage from the given lines, "These climatic shifts will have an impact on the hydrological cycles in the region, lead to more run-off and less recharge, and affect the ground water tables." Can be stated that the climate changes increase the criticality of water resources while simultaneously threatening it is the main crux of the passage. So, option (c) is correct and rest of all three are incorrect.

Passage–4

Researchers were able to use stem cells to gauge the neurotoxic effects of the environmental pollutant Bisphenol A (BPA). They used a combination of biochemical and cell-based assays to examine the gene expression profile during the differentiation of mouse embryonic stem cells upon treatment with BPA, a compound known to cause heart diseases, diabetes, and developmental abnormalities in humans. They were able to detect and measure BPA toxicity towards the proper specification of primary germ layers, such as endoderm and ectoderm, and the establishment of neural progenitor cells.

78. On the basis of the passage given above, the following assumptions have been made:

1. BPA may alter embryonic developments in vivo.
2. Biochemical and cell-based assays are useful in finding out treatments for pollution-induced diseases.
3. Embryonic stem cells could serve as a model to evaluate the physiological effects of environmental pollutants.

Which of the above assumptions are valid?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c) : According to the passage the Bisphenal A (BPA). May alter embryonic development in vivo from the line "mouse embryonic stem cells upon treatment with BPA, a compound known to cause heart diseases, diabetes, and developmental abnormalities in humans". So it can be clearly assumed statement 1 is correct. Statement 3 is correct because from the passage that embryonic stem cells could serve as a model to evaluate the physiological effect of environmental pollutants.
Hence, option (c) is correct.

Direction for the following 4 (Four) item :

Read the Following four passages and answer the items that follow. Your answers to these items should be based on the passages only.

Passage – 1

Nothing can exist in a natural state which can be called good or bad by common assent, since every man who is in a natural state consults only his own advantage, and determines what is good or bad according to his own fancy and insofar as he has regard for his own advantage alone, and holds himself responsible to no one save himself by any law; and therefore sin cannot be conceived in a natural state, but only in a civil state, which is decreed by common consent what is good or bad, and each one holds himself responsible to the state.

79. Which one of the following statement best reflects the central idea of the passage given above?

(a) The conceptions of what is right or wrong exist due to the formation of a state.
(b) Unless a ruling authority decides as to what is right or wrong, no man would be morally right.
(c) Man is inherently immoral and selfish in a natural state.
(d) The idea of what is right or wrong is necessary for the survival of human species.

Ans. (a) : The conceptions of what is Right or what is wrong exist due to the formation of a state, is justified from the lines, "sin cannot be conceived is a natural state, but only a civil state, which is decreed by common consent what is good or bad, and each one holds himself responsible to the sate". So option (a) is right. Reset of all are incorrect.

80. Which one of the following habits is found more often in good people ?

(a) Mixing up the true and false
(b) Intentional mixing up of truth with the false
(c) Falsification of facts
(d) Complete concealment of truth

Ans. (b) : According to the passage the most appropriate answer will be option (b). The second half of the passage mentions that good people often tell lies which contains some kind of truth. Hence there is an intentional mix up of truth and false. Another options are inappropriate.
Hence, option (b) is correct.